

Table 1. Sensitivity analysis of the hyperparameter α . Sensitivity is measured via the Coefficient of Variation ($CV = \sigma/\mu$) to ensure a fair comparison across datasets with varying base performance.

Dataset	$\alpha = 0.1$	$\alpha = 0.5$	$\alpha = 0.9$	$\alpha = 1.5$	$\alpha = 2.0$	Sensitivity (CV)
MATH	74.29	74.26	74.40	74.55	74.38	0.14%
GSM8K	87.58	87.71	87.64	87.76	87.68	0.07%
Olympiad	35.25	35.12	35.16	35.04	35.19	0.20%
OMNI-MATH	16.52	16.59	16.54	16.59	16.55	0.17%
Average						0.14%

Table 2. Sensitivity analysis of the hyperparameter β . Sensitivity is measured via the Coefficient of Variation ($CV = \sigma/\mu$) to ensure a fair comparison across datasets with varying base performance.

Dataset	$\beta = 0.1$	$\beta = 0.5$	$\beta = 1.0$	$\beta = 1.5$	$\beta = 2.0$	Sensitivity (CV)
MATH	74.36	74.40	74.41	74.35	74.43	0.04%
GSM8K	87.67	87.64	87.60	87.64	87.59	0.03%
Olympiad	35.24	35.16	35.14	35.17	35.20	0.10%
OMNI-MATH	16.49	16.54	16.52	16.58	16.46	0.25%
Average						0.11%

Table 3. Sensitivity analysis of the hyperparameter γ in the proposed VFPO. Sensitivity is measured via the Coefficient of Variation ($CV = \sigma/\mu$) to ensure a fair comparison across datasets with varying base performance.

Dataset	$\gamma = 0.005$	$\gamma = 0.02$	$\gamma = 0.1$	$\gamma = 0.15$	$\gamma = 0.5$	Sensitivity (CV)
MATH	74.33	74.40	74.42	74.35	74.38	0.04%
GSM8K	87.57	87.64	87.52	87.62	87.66	0.06%
Olympiad	35.13	35.16	35.24	35.18	35.15	0.11%
OMNI-MATH	16.51	16.54	16.48	16.49	16.56	0.18%
Average						0.10%

Table 4. Sensitivity analysis of the hyperparameter λ^- in the proposed VFPO. Sensitivity is measured via the Coefficient of Variation ($CV = \sigma/\mu$) to ensure a fair comparison across datasets with varying base performance.

Dataset	$\lambda^- = 0.1$	$\lambda^- = 0.5$	$\lambda^- = 0.9$	$\lambda^- = 1.5$	$\lambda^- = 2.0$	Sensitivity (CV)
MATH	74.37	74.41	74.40	74.35	74.46	0.05%
GSM8K	87.62	87.55	87.64	87.68	87.61	0.05%
Olympiad	35.14	35.19	35.16	35.08	35.14	0.10%
OMNI-MATH	16.47	16.55	16.54	16.58	16.55	0.22%
Average						0.11%

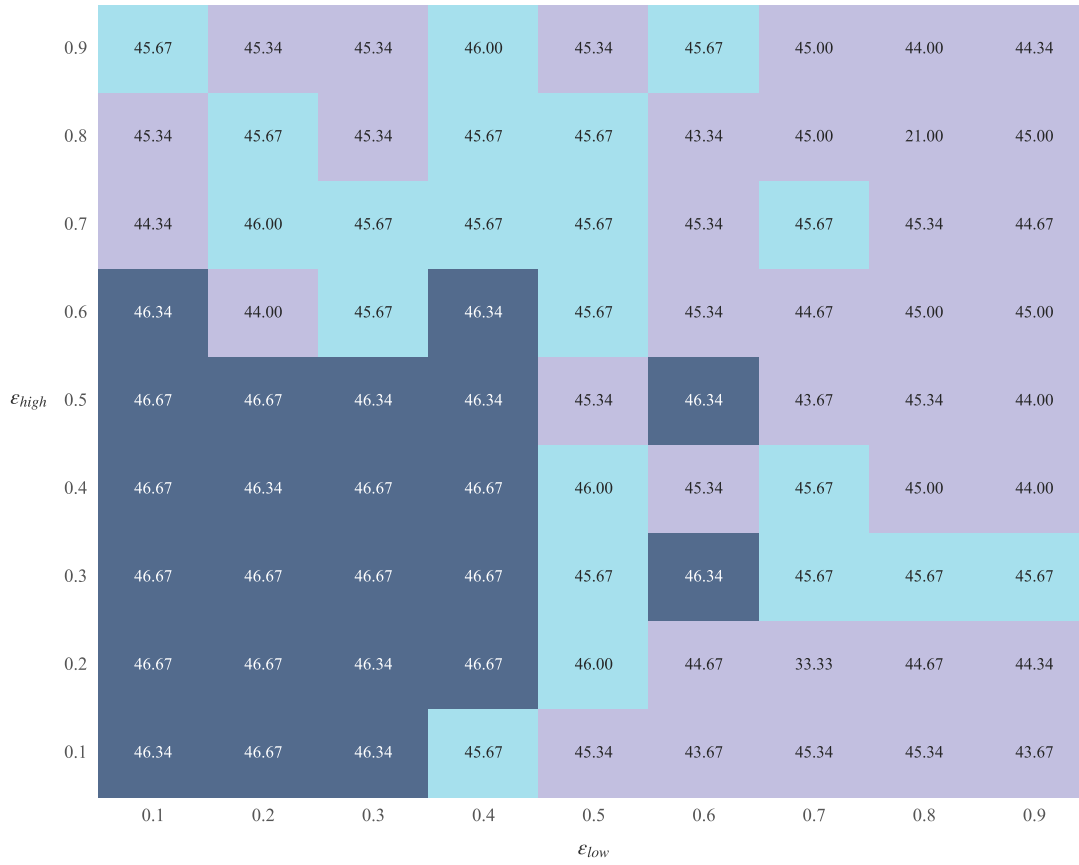


Figure 1. sensitivity analysis of ϵ_{low} and ϵ_{high} . The heatmap reveals a **broad and flat optimal plateau** (notably in the lower-left region), where the model maintains consistently high performance with minimal variance.

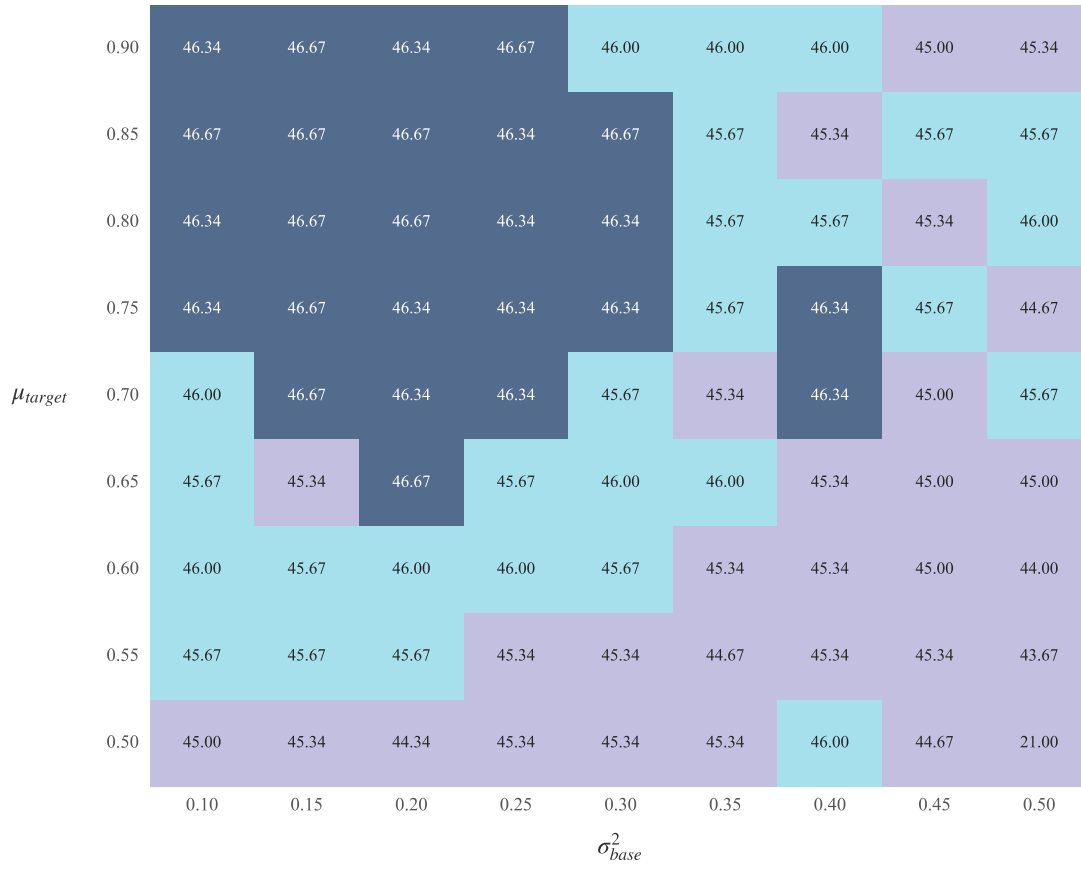


Figure 2. **Sensitivity analysis** of σ_{base}^2 and μ_{target} . The heatmap illustrates a **substantial and stable peak performance region**, particularly as both parameters increase (top-left quadrant).